Data Analytics Project 2

[Exploratory Data Analysis Report]

Introduction

This report presents the findings of an Exploratory Data Analysis (EDA) conducted on a cleaned e-commerce sales dataset. The analysis was performed in Python using pandas, NumPy, Matplotlib and Seaborn and covers four major analytical dimensions: descriptive statistics, distribution analysis, trend and pattern analysis, and correlation analysis.

The dataset used is the output of Project 1 (data cleaning) and contains transaction-level records including product, pricing, date, order status, and referral source information.

Descriptive Statistics

Summary statistics were computed for all numeric columns using df.describe(). A custom table was additionally printed showing the mean, median and skewness for each column, with a '(skewed)' flag applied when the absolute skewness exceeded 1.0.

Key Metrics Captured

- Count, mean, standard deviation, minimum and maximum values.
- 25th, 50th (median) and 75th percentile values.
- Skewness coefficient - values above |1| indicate a significantly skewed distribution.

Skewness analysis is particularly important for revenue-related columns like TotalPrice, where extreme transactions (very large or very small orders) can pull the mean away from the median, signalling a non-normal distribution.

DESCRIPTIVE STATISTICS FOR NUMERICAL COLUMNS

	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200.00	1200.00	1200.00	1200.00
mean	2.95	356.41	5.48	1053.97
std	1.41	197.18	2.28	819.86
min	1.00	11.39	1.00	11.39
25%	2.00	186.06	4.00	410.52
50%	3.00	364.21	5.00	823.62
75%	4.00	521.57	7.00	1578.48
max	5.00	699.93	10.00	3456.40

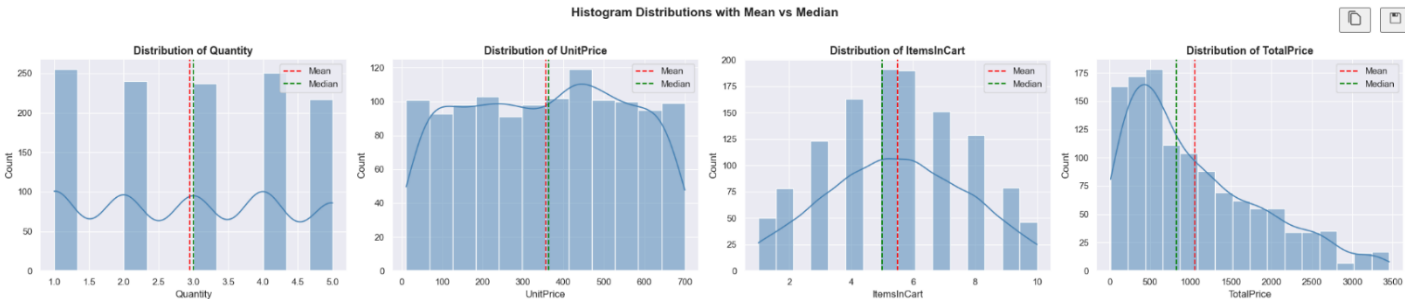
Column	Mean	Median	Skew
Quantity	2.95	3.00	0.03
UnitPrice	356.41	364.21	-0.03
ItemsInCart	5.49	5.00	0.00
TotalPrice	1053.97	823.62	0.89

Distribution Analysis

Distribution plots were generated for all numerical columns to visually inspect the shape and spread of the data.

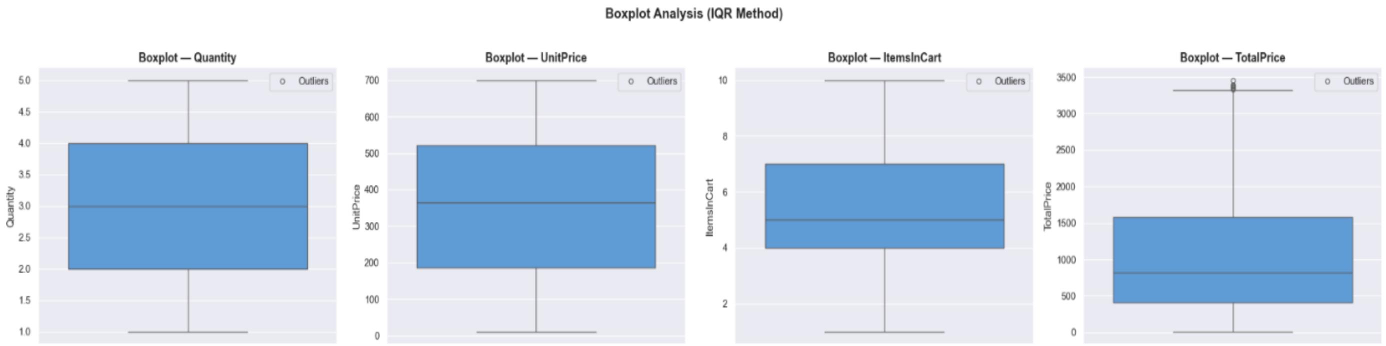
Histograms with KDE

Kernel Density Estimate (KDE) histograms were produced for each numeric column. Red dashed lines mark the mean and green dashed lines mark the median, making it straightforward to see the direction and degree of skew at a glance.



Boxplots (IQR Method)

Boxplots display the interquartile range (IQR), median and outlier positions for each numeric variable. Outlier points beyond the whiskers (Q1 - 1.5×IQR and Q3 + 1.5×IQR) are rendered in red for visual clarity.



IQR Outlier Detection – Summary

For each numeric column the following were computed programmatically:

```
IQR OUTLIER DETECTION
Quantity
Q1=2.00  Q3=4.00  IQR=2.00
Fences: [-1.00, 7.00]
Outliers detected: 0

UnitPrice
Q1=186.06  Q3=521.57  IQR=335.51
Fences: [-317.20, 1024.83]
Outliers detected: 0

ItemsInCart
Q1=4.00  Q3=7.00  IQR=3.00
Fences: [-0.50, 11.50]
Outliers detected: 0

TotalPrice
Q1=410.52  Q3=1578.47  IQR=1167.95
Fences: [-1341.41, 3330.41]
Outliers detected: 8
```

Trend & Pattern Analysis

The Date column was parsed to datetime format, and the following time features were extracted for trend analysis: Month (numeric), Month Name (abbreviated string), Year and Day of Week.

Monthly Revenue Trend

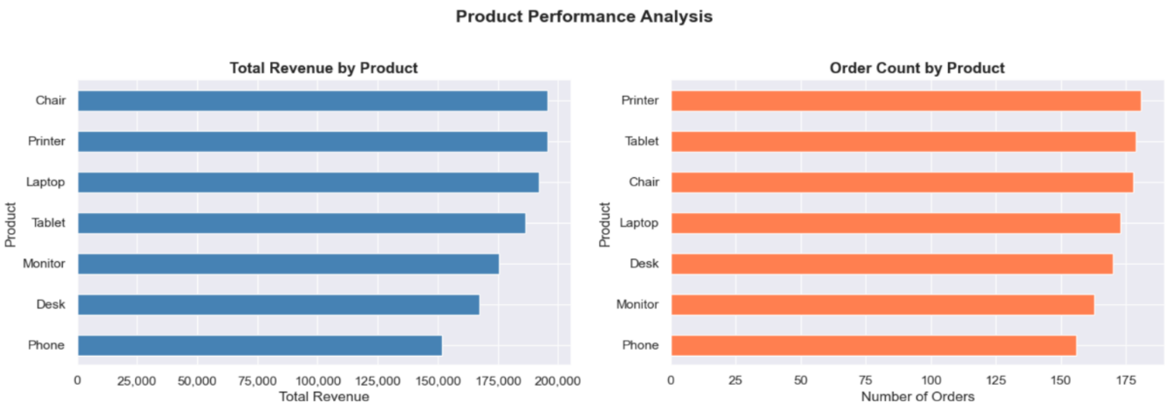
Revenue was aggregated by year-month period to construct a line chart showing the revenue trajectory over the full date range. An area fill was added beneath the line to emphasize the magnitude of revenue at each period. This chart is the primary tool for identifying seasonal peaks, growth phases and revenue dips.

Product Performance

Two side-by-side horizontal bar charts were produced:

- Total Revenue by Product - identifies the highest-grossing products
- Order Count by Product - identifies the most frequently ordered products

Comparing these two charts reveals whether high-revenue products are also high-volume (indicating consistent demand) or whether a small number of large orders drive their revenue figure.



Referral Source Performance

Performance was broken down by acquisition channel (ReferralSource) across three KPIs:

```
Referral Source Performance:
                Total Revenue  Order Count  Avg Order Value
ReferralSource
Instagram           275285.45          259          1062.88
Email               261808.55          250          1047.23
Google              250441.48          241          1039.18
Facebook            250410.90          228          1098.29
Referral            226815.58          222          1021.69
```

Correlation Analysis

Pearson Correlation Heatmap

A Pearson correlation matrix was computed across all numeric columns and displayed as a lower-triangle heatmap (to avoid redundancy). The coolwarm diverging palette maps values from -1 (strong negative correlation) through 0 (no linear relationship) to +1 (strong positive correlation). Annotated cell values enable precise reading of each coefficient.

Product vs Order Status Pivot Heatmap

A pivot table was constructed showing the average TotalPrice for each combination of Product and OrderStatus. The resulting heatmap (YlGnBu palette) reveals which product-status combinations generate the highest average revenue; for example, whether completed orders for a specific product significantly outperform cancelled orders in monetary terms.